



សាកលវិទ្យាល័យ ប៊ែលទី អន្តរជាតិ

BELTEI INTERNATIONAL UNIVERSITY

គុណភាព ប្រសិទ្ធភាព ឧត្តមភាព សីលធម៌ គុណធម៌ Quality Efficiency Excellence Morality Virtue

The Future of Global Leaders

មហាវិទ្យាល័យ ព័ត៌មានវិទ្យា និង វិទ្យាសាស្ត្រ

FACULTY OF INFORMATION TECHNOLOGY AND SCIENCE

MAJOR: SOFTWARE ENGINEERING

SUBJECT: SOFTWARE PROJECT MANAGEMENT

TOPIC: "AI TRAVEL GUIDE CHATBOT FOR CAMBODIA"

FINAL PROJECT

LECTURER PROFESSOR

DR. SOVANN CHEN



សាកលវិទ្យាល័យ ប៊ែលទី អន្តរជាតិ

BELTEI INTERNATIONAL UNIVERSITY

គុណភាព ប្រសិទ្ធភាព ឧត្តមភាព សីលធម៌ គុណធម៌ Quality Efficiency Excellence Morality Virtue

The Future of Global Leaders

SUBMITTED BY GROUP 4



Mr. TOB CHENDA



Mr. TEP VANNY



Mr. VAN SARATH



Mr. PHAN CHHAYLEAP



Mr. YI PHANIT



សាកលវិទ្យាល័យ ប៊ែលទី អន្តរជាតិ

BELTEI INTERNATIONAL UNIVERSITY

គុណភាព ប្រសិទ្ធភាព ឧត្តមភាព សីលធម៌ គុណធម៌ Quality Efficiency Excellence Morality Virtue

The Future of Global Leaders



Mr. TEP VANNY
Marketing Officer



Mr. VAN SARATH
UI/UX Designer
Frontend Developer



Mr. TOB CHENDA
Project Manager
& Developer



Mr. PHAN CHHAYLEAP
System Analyst
Backend Developer



Mr. YI PHANIT
Tester (QA)

“AI TRAVEL GUIDE CHATBOT FOR CAMBODIA”

25-December- 2025

 vannymanuel333666999@gmail.com



CONTENT

01

PROBLEM STATEMENT

07

TESTING RESULTS

02

PROPOSED SOLUTION

08

CHALLENGES & SOLUTIONS

03

KEY FEATURES

09

CONCLUSION & FUTURE SCOPE

04

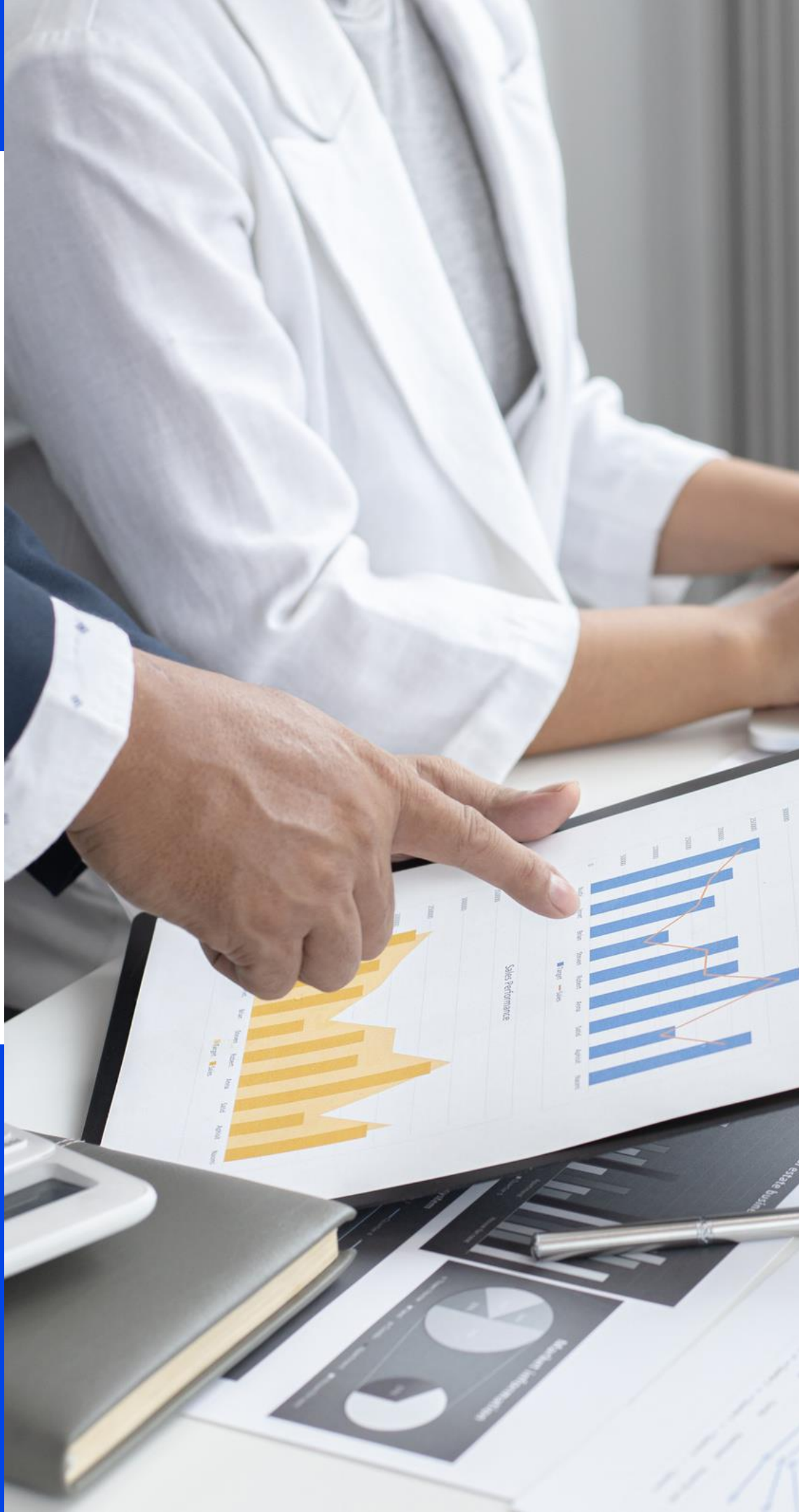
TECHNOLOGY STACK

05

SYSTEM ARCHITECTURE DIAGRAM

06

DEMO





1. PROBLEM STATEMENT (WHY DID WE BUILD THIS?)

WHY DID WE BUILD THIS?

- Tourists visiting Cambodia face difficulties finding accurate, reliable, and localized travel information
- Existing travel platforms are:
 - ❖ Difficult to navigate, with scattered information across multiple pages.
 - ❖ Not focused specifically on Cambodia, lacking deep local insights.
- International tourists experience language barriers, making it hard to understand local guidance.
- There is no 24/7 intelligent assistant to instantly answer tourist questions.

As a result, tourists spend more time searching for information and may miss important travel opportunities





2. PROPOSED SOLUTION (High-level overview)

This project is an AI-powered travel guide chatbot that assists domestic and international tourists with travel information by:

- ❑ Recommending places in Cambodia (Angkor Wat, Phnom Penh, Siem Reap, beaches, cultural heritage sites, and natural attractions, etc.).
- ❑ Giving travel tips (transport, food, culture, weather, safety)
- ❑ Answer tourists' questions using English (and optionally Khmer)
- ❑ Provide 24/7 instant responses to users
- ❑ Improve tourist experience and satisfaction
- ❑ Accessible via web or mobile application
- ❑ Easy-to-use and user-friendly interface
- ❑ Promote Cambodia's tourism industry using modern AI technology

To address these problems, this project proposes the development of an AI Travel Guide Chatbot for Cambodia that provides instant, accurate, and personalized travel information using Artificial Intelligence.



3. KEY FEATURES

In-Scope Features

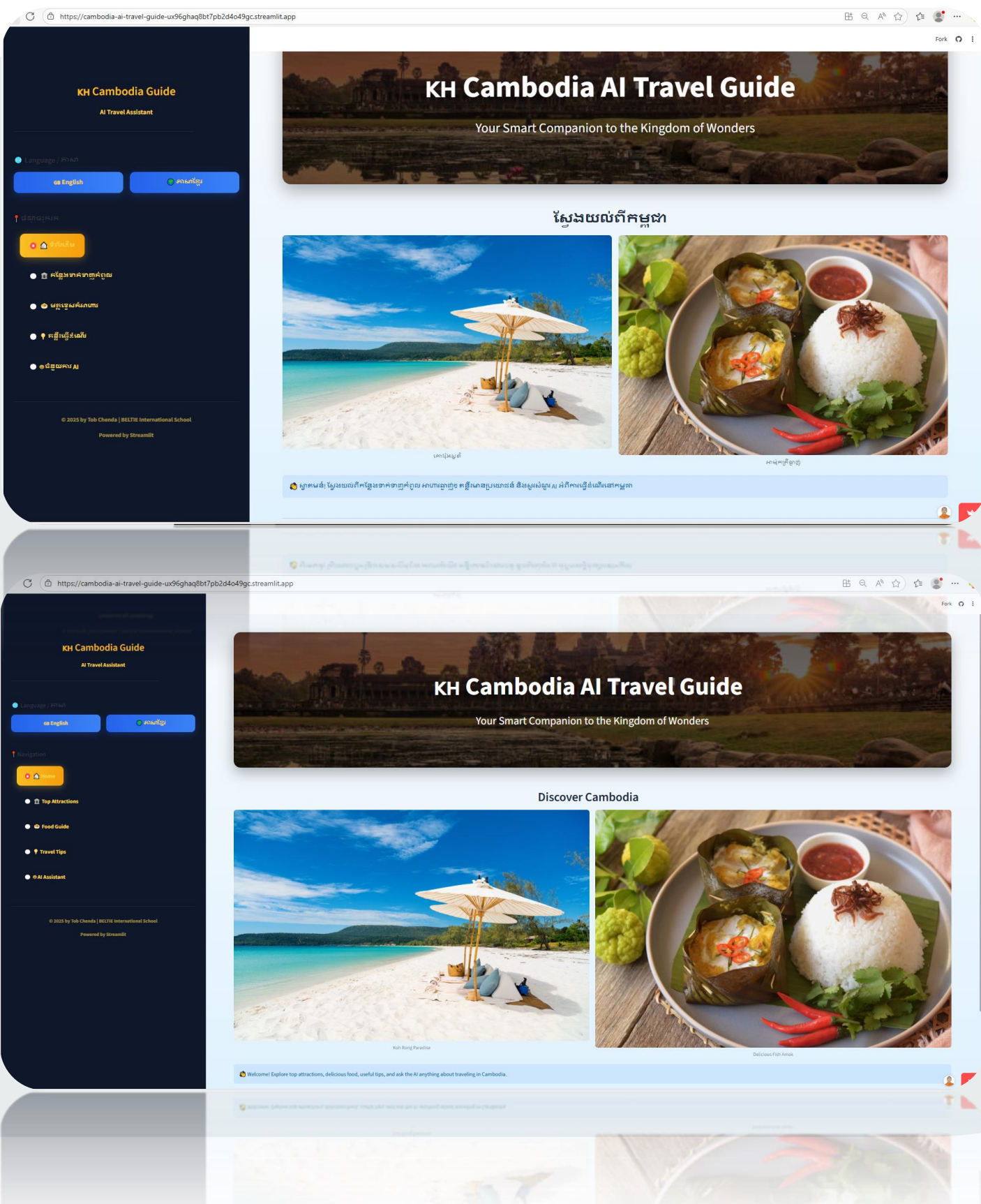
1. Language Support

- ✓ English
- ✓ Khmer

Enables communication for both international and local tourists
Helps reduce language barriers

2. Home Page

- ✓ Simple and user-friendly interface
- ✓ Quick access to main features
- ✓ Highlights popular destinations and chatbot access





3. KEY FEATURES

3. Tourist Top Attractions

Recommends popular places in Cambodia, including:

- Angkor Wat
- Phnom Penh
- Siem Reap
- Beaches (Sihanoukville, Koh Rong, etc.)
- Cultural heritage sites
- Natural attractions

Provides brief descriptions and travel information

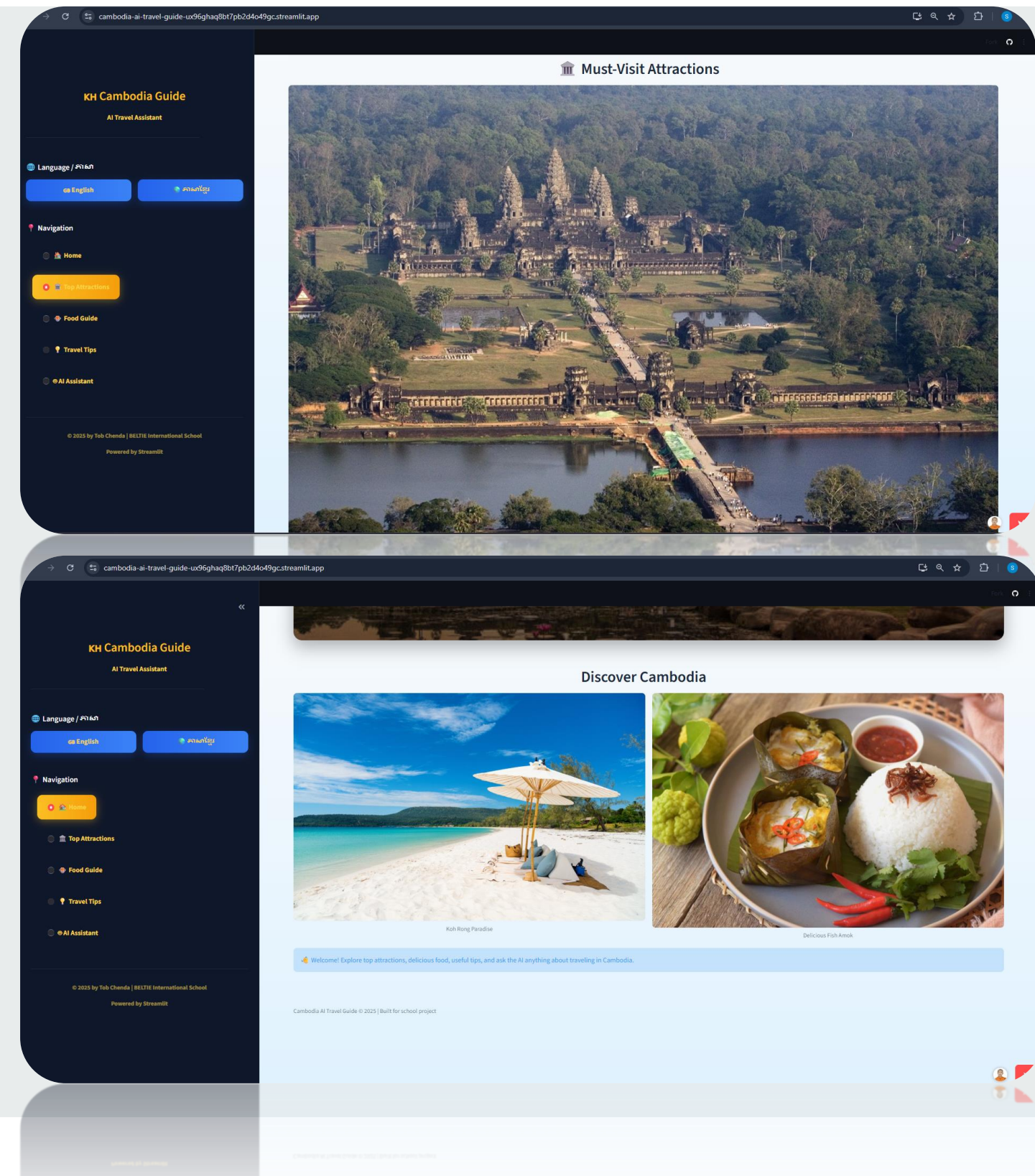
4. Food Guide

Suggests popular Cambodian foods:

- Amok
- Lok Lak
- Bai Sach Chrouk
- Street food and walk street recommendations

Includes food descriptions and where to try them

Helps tourists explore local cuisine safely





3. KEY FEATURES

5. Travel Tips

Transportation guidance

Weather and best travel seasons

Cultural practices and etiquette

Safety tips for tourists

Currency and basic local information

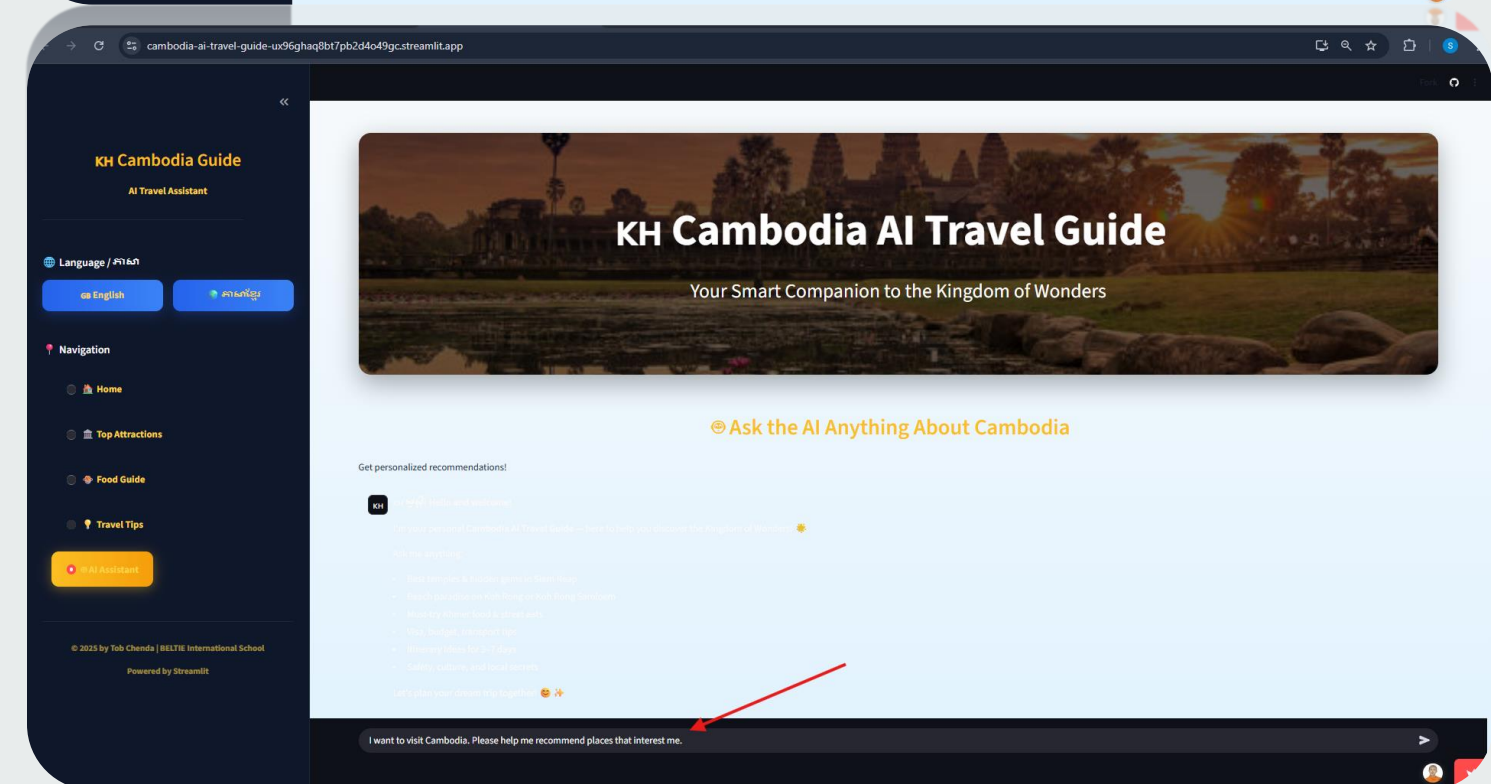
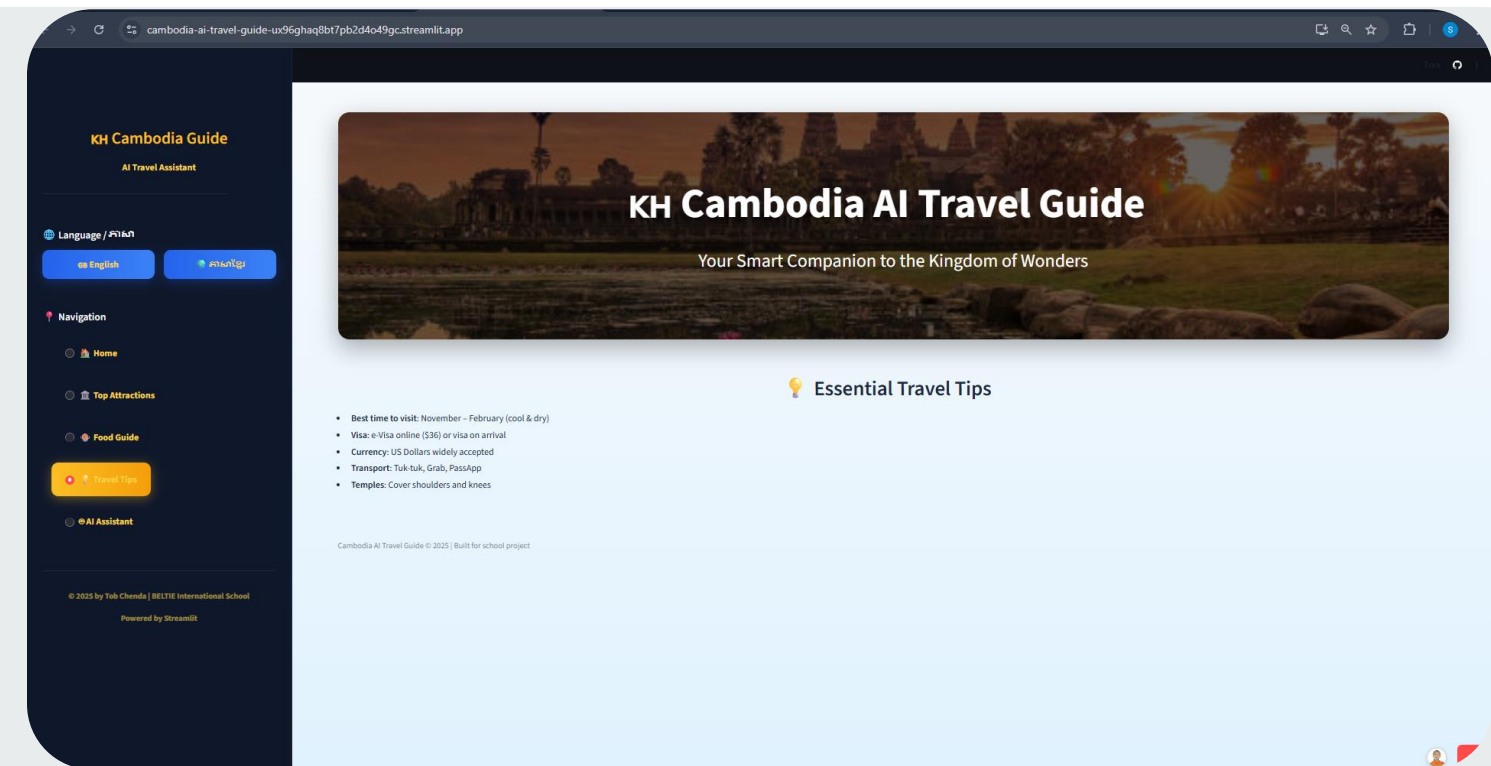
6. AI Assistance

AI-powered chatbot available 24/7

Answers travel-related questions instantly

Provides personalized recommendations

Supports natural language conversation





4. TECHNOLOGY STACK

Frontend

HTML CSS

Used to build a responsive and interactive user interface for the chatbot.

Backend

Django (Python)

Handles API requests, chatbot logic, and communication with the AI service.

Database

database: PostgreSQL



SQL

Stores travel information, user queries, chat logs, and system data.

AI / NLP Component



AI / NLP Engine

Processes user input and generates intelligent responses.

Development Tools



VS Code – Code editor



Git & GitHub – Version control



5. SYSTEM ARCHITECTURE DIAGRAM

Components

1. User (Tourist)

- Accesses the system via **Web or Mobile Browser**
- Interacts using **English or Khmer**

2. Frontend (React)

- Provides user interface:
 - Home
 - Tourist Attractions
 - Food Guide
 - AI Chatbot
- Sends user requests to backend via **API**

3. Backend Server (Django)

- Handles business logic
- Processes user requests
- Manages communication between:
 - Frontend
 - AI / NLP Engine
 - Database

4. AI / NLP Engine

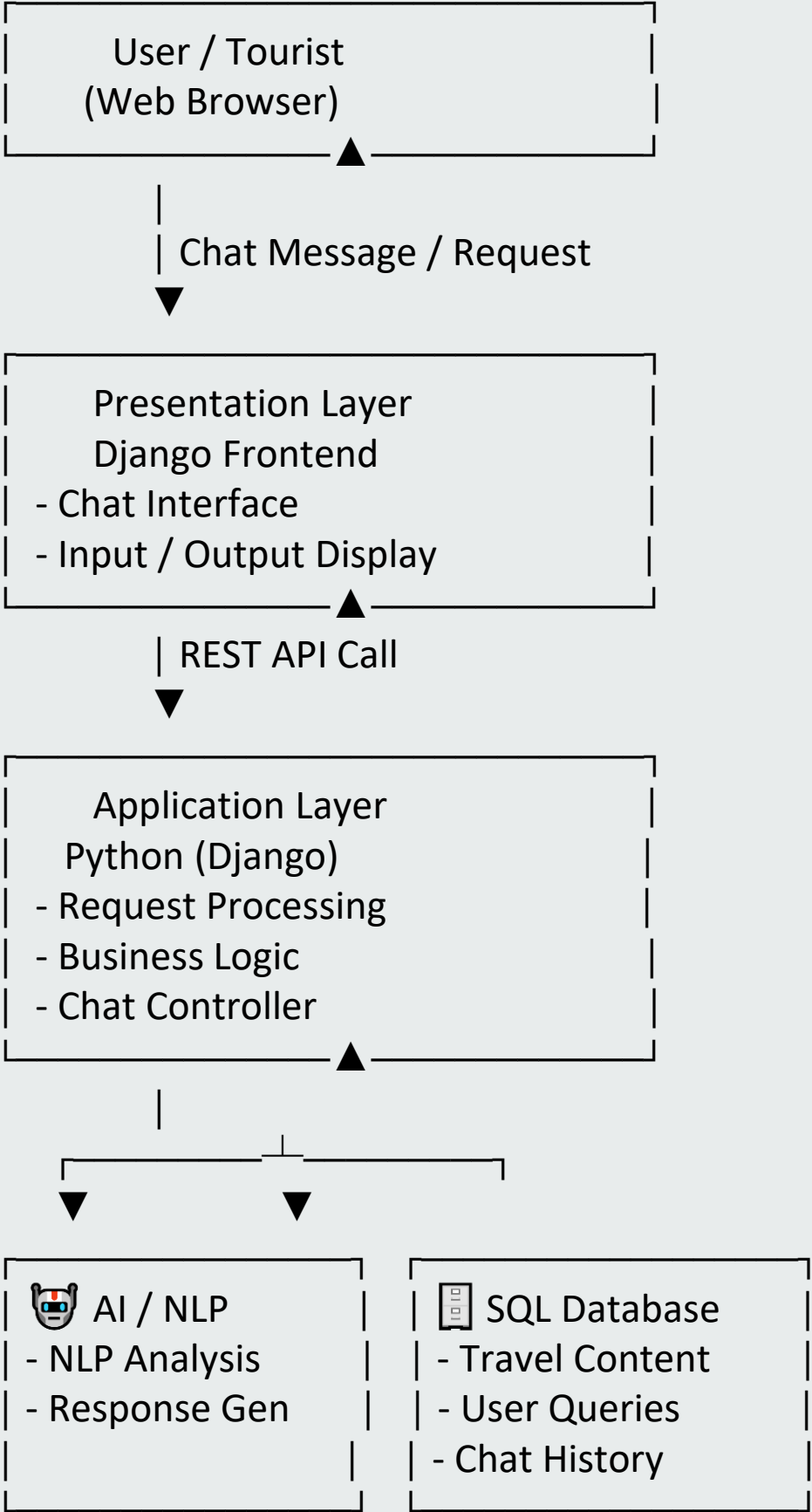
- Understands user queries
- Generates intelligent responses and recommendations
- Supports **multilingual interaction (English–Khmer)**

5. Database (SQL)

- Stores:
 - Tourist attractions
 - Food guides
 - Travel tips
 - User preferences
 - Chatbot conversation logs



5. SYSTEM ARCHITECTURE DIAGRAM





6. DEMO (Switch to live software demo - *Most Important Part*)

Demo Objective

- Demonstrate how the **AI Travel Guide Chatbot for Cambodia** works in real time.
- Show the system's **usability, accuracy, and responsiveness**.

Demo Flow (What We Will Show)

1. Open the chatbot web application.
2. Show the **clean chat interface**.
3. Enter sample tourist questions.
4. Display **AI-generated responses instantly**.
5. Highlight Cambodia-specific information.

Sample Demo Questions

- “What are the best places to visit in Siem Reap?”
- “How can I travel from Phnom Penh to Sihanoukville?”
- “What local food should I try in Cambodia?”
- “Is Cambodia safe for tourists?”

Key Points to Highlight During Demo

- Fast response time
- Accurate and relevant answers
- Interactive chat experience
- User-friendly design



7. Testing Result (Result of Project)

Deploy

KH Cambodia AI Travel Guide

Login or Register

Login

Register

Username

Email

Password

Register

English

ខ្មែរ

Home

Top Attractions

Food Guide

Travel Tips

AI Assistant

Deploy

KH Cambodia AI Travel Guide

Welcome, Phan ChhayLeap

Ask the AI Anything About Cambodia

Hello

Hello! How can I assist you today? Are you planning a trip to Cambodia or do you have any specific questions about the country?

best place to travel in Cambodia during winter?

Cambodia is a wonderful destination during the winter months (November to February) when the weather is cool and dry. Here are some of the best places to visit during this

Home

Top Attractions

Food Guide

Travel Tips

AI Assistant

Logout

Perched on a mountaintop, this temple offers breathtaking views and is less crowded than Angkor Wat, making it a serene place to explore.

Remember to pack light layers, as the weather can be cool in the evenings! If you need more specific recommendations on activities or accommodations, feel free to ask! Enjoy your travels in Cambodia!

Ask me anything (e.g., 'Plan a 5-day trip', 'Best beaches', 'Vegetarian food options'):

© 2025 Cambodia AI Travel Guide | Powered by Streamlit & OpenAI



8. CHALLENGES & SOLUTIONS (TECHNICAL HURDLES FACED)

Challenges

- ❑ Handling diverse user questions
- ❑ Ensuring accurate tourism information
- ❑ AI response consistency
- ❑ Language understanding

Solutions

- ✓ Improved NLP training
- ✓ Structured tourist data
- ✓ Continuous testing and refinement
- ✓ User feedback integration

9. CONCLUSION & FUTURE SCOPE

CONCLUSION

The **AI Travel Guide Chatbot for Cambodia** was successfully designed and developed. The system provides **instant, accurate, and personalized travel information** to tourists. It supports **English and Khmer**, helping overcome language barriers. The chatbot enhances **tourist experience and satisfaction** through 24/7 AI assistance. The project demonstrates how **modern AI technology** can support and promote Cambodia's tourism industry.

9. CONCLUSION & FUTURE SCOPE

Future Scope

- **Mobile Application Development**
Android and iOS versions for wider accessibility
- **Voice-Based Chatbot**
Support voice input and output for hands-free interaction
- **Advanced Personalization**
AI recommendations based on user behavior, interests, and travel history
- **Map & Location Integration**
Integration with GPS and maps for navigation and nearby attractions
- **Booking & Service Integration**
Hotels, transportation, tours, and ticket booking
- **Real-Time Information**
Live updates on weather, events, traffic, and travel advisories
- **Expanded Language Support**
Improved Khmer support and additional international languages

THANK YOU

FOR YOUR ATTENTION

25 – December- 2025



vannymanuel333666999@gmail.com

